Computer Programming 1

Lab 1 – starting exercises

(16-09-2026)

Terminal (Shell) – Task 01

Tasks

1. Open a terminal,
2. Navigate the filesystem till reaching the folder “Desktop”,
3. Obtain the list of elements in the current folder,
4. Close the terminal

Terminal (Shell) – Task 02

Tasks

1. Open the terminal,
2. Create a folder named “test”
3. Enter in the created folder
4. Create an empty file
5. Turn back to the starting folder (parent of “test”)
6. Check if the created folder (named “test”) still exists
7. Remove the created folder (named “test”)
8. Check again if the created folder (named “test”) exists
9. Close the terminal

Programming – Task 03 *(First program)*

Task

1. Write a simple C++ program with empty main
2. Compile it
3. Run it

Note:

Alt Gr + 7 → {

Alt Gr + 0 → }

Implementation constraint: use only a Text editor, no VS Code or any other IDEs

Programming – Task 03 (*Solution*)

1. Open a Terminal
2. Start a text editor
   - open Text Editor and creates a file named “example.cpp”
3. In the editor, write the program code
4. In the terminal, compile the program
   - “g++ example.cpp”
5. In the terminal, execute the program
   - ./a.out

Example program:

```
int main()
{
   return 0;
}
```

Exercises

Programming – Task 04a *(Second program)*

Task -- Write a simple C++ program that:

1. Defines two variables in the main function: an integer variable and a char variable
2. Initialize both variables with some valid values by using 12 and ‘A’
3. Print the values of both variables to the video, like:

   var1 value1

   var2 value2

Implementation constraint: use only a Text editor, no VS Code or any other IDEs

Programming – Task 04b

Task -- Write a simple C++ program that:

1. Defines two variables in the main function: an integer variable and a char variable
2. Initialize both variables with some valid values by using 12 and ‘A’
3. Print the values of both variables to the video

Implementation constraint: use only a Text editor, no VS Code or any other IDEs

Implementation variant #1: after step3, change the value of the variables and print them again

Programming – Task 04c

Task -- Write a simple C++ program that:

1. Defines two variables in the main function: an integer variable and a char variable
2. Initialize both variables with some valid values by using 12 and ‘A’
3. Print the values of both variables to the video

Implementation constraint: use only a Text editor, no VS Code or any other IDEs

Implementation variant #2: after the implementation of variant #1, remove the variables initialization and re-compile and execute the program

Programming – Task 05 *(it needs output stream)*

Task -- Write a simple C++ program that:

1. Takes one input character from the keyboard
2. Stores the values of the input in a variable
3. Print the vale of the variable to the video

Implementation constraint: use only a Text editor, no VS Code or any other IDEs

Solutions

Programming – Task 04a *(Solution)*

Example program:

```
1.  #include <iostream>
2.  using namespace std;
3.
4.  int main(){
5.   int x=12;
6.   char y='A';
6.   cout << " Print the value of x: " << x << endl;
7.   cout << " Print the value of y: " << y << endl;
9.   return 0;
10. }
```

Programming – Task 04b *(Solution)*

Example program:

```
1.  #include <iostream>
2.  using namespace std;
3.
4.  int main(){
5.   int x=12;
6.   char y='A';
6.   cout << " Print the value of x: " << x << endl;
7.   cout << " Print the value of y: " << y << endl;
8.   x=14;
9.   y='B';
10.  cout << " Print the value of x: " << x << endl;
11.  cout << " Print the value of y: " << y << endl;
12.  return 0;
10. }
```

Programming – Task 04c *(Solution)*

Example program:

```
1.  #include <iostream>
2.  using namespace std;
3.
4.  int main(){
5.   int x=12;
6.   char y='A';
6.   cout << " Print the value of x: " << x << endl;
7.   cout << " Print the value of y: " << y << endl;
8.   x=12;
9.   y='B';
10.  cout << " Print the value of x: " << x << endl;
11.  cout << " Print the value of y: " << y << endl;
12.  return 0;
10. }
```

Programming – Task 05 *(Solution)*

Example program:

```
1.  #include <iostream>
2.  using namespace std;
3.
4.  int main(){
5.   char character;
6.   cout << "Please, insert a character: ";
7.   cin >> character;
8.   cout << "The inserted character is: " << character << endl;
9.   return 0;
10. }
```